CHANNABASAVESHWARA INSTITUTE OF TECHNOLOGY

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RATTLE BOTS CLUB PRESENTS

SCAN TO REGISTER

CAMPUS LOCATION

ROBOMANIA 2K26

NATIONAL LEVEL ROBOTICS COMPETITION

APRIL 2026

13TH - MONDAY

FROM DESIGN TO DOMINATION

ROBO-RACE

PROBLEM STATEMENT:

PARTICIPANTS MUST BUILD AND OPERATE A MANUALLY CONTROLLED ROBOT, EITHER WIRED OR WIRELESS, TO NAVIGATE A TRACK FILLED WITH OBSTACLES AND SHARP TURNS IN THE SHORTEST POSSIBLE TIME.

GAMEPLAY:

- THE ROBOT BEGINS AT THE STARTING LINE, AND THE TIMER STARTS AS SOON AS THE RACE BEGINS.
- THE ROBOT MUST STAY ON TRACK; IF IT MOVES OFF-COURSE, IT MUST RESTART FROM THE LAST CROSSED CHECKPOINT WHILE THE TIMER CONTINUES RUNNING.
- SKIPPING AN OBSTACLE INCREASES 30 SECONDS PENALTY POINTS, WHICH ARE ADDED TO THE TOTAL TIME.

PENALTIES:

- 5 SECONDS ADDED IF ONE WHEEL MOVES OUT OF THE TRACK.
- 20 SECONDS ADDED IF THE ENTIRE ROBOT MOVES OUT OF THE TRACK.
- 30 SECONDS ADDED FOR MANUAL INTERVENTION (HAND TOUCH).
- THE RACE ENDS WHEN THE ROBOT CROSSES THE FINISH LINE, AT WHICH POINT THE TIMER STOPS.
- SCORING AND PENALTY DETAILS WILL BE DISCLOSED ON THE DAY OF THE EVENT.
- IF BOT STOPS FUNCTIONING FOR MORE THAN 3 MINUTES WILL BE DISQUALIFIED

ROBOT SPECIFICATIONS:

- THE ROBOT CAN BE WIRED OR WIRELESS BUT MUST BE MANUALLY CONTROLLED.
- MAXIMUM DIMENSIONS: 300MM X 300MM X 300MM (L X B X H). ANY BOT EXCEEDING THIS LIMIT WILL BE DISQUALIFIED.(0 TOLERANCE)
- MAXIMUM WEIGHT: 5 KG.
- THE BOT MUST NOT BE BUILT FROM LEGO PARTS OR PRE-ASSEMBLED KITS. IT MUST BE ELECTRICALLY POWERED—IC ENGINES ARE PROHIBITED.
- POWER CAN BE SUPPLIED VIA ONBOARD BATTERIES OR A STATIONARY
- POWER SOURCE CONNECTED VIA A WIRE.
- MAXIMUM VOLTAGE: 12V ACROSS ANY TWO POINTS OF THE ROBOT.
- TEAMS SHOULD CARRY SPARE BATTERIES TO AVOID DISQUALIFICATION DUE TO POWER FAILURE.
- IF USING A WIRED CONNECTION, THE WIRE MUST BE LONG ENOUGH TO COVER THE RACE TRACK AND REMAIN SLACK AT ALL TIMES.

ROBO-RACE

RACE TRACK SPECIFICATIONS:

- THE TRACK WILL HAVE A SINGLE START AND FINISH LINE. THE SURFACE MAY HAVE IRREGULARITIES.
- VARIOUS OBSTACLES WILL BE PRESENT TO SLOW DOWN THE ROBOT'S MOVEMENT

RULES AND REGULATIONS:

- EACH TEAM CAN HAVE UPTO 4 MEMBERS.
- TWO TEAM MEMBERS WILL BE DESIGNATED AS THE ROBOT HANDLER, RESPONSIBLE FOR CONTROLLING THE ROBOT DURING THE GAME.
- OTHER TEAM MEMBERS MUST STAY OUTSIDE THE COMPETITION ZONE.
- MISBEHAVIOR BY ANY PARTICIPANT MAY RESULT IN EXPULSION OR DISQUALIFICATION.
- DELIBERATE INTERFERENCE WITH OTHER ROBOTS OR DAMAGING THE ARENA WILL RESULT IN DISQUALIFICATION.
- ALL DECISIONS REGARDING SCORING, GAMEPLAY, TIMING, AND PARTICIPATION MADE BY THE ORGANIZING COMMITTEE ARE FINAL.
- TEAMS MUST RESPECT THE REFEREE'S CALLS AND JUDGMENTS. ORGANIZERS MAY MODIFY RULES ON SPOT

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STUDENT COORDINATOR

ROBO SOCCER

OBJECTIVE:

TEAMS COMPETE TO SCORE GOALS BY MANEUVERING A BALL INTO THE OPPONENT'S NET USING THEIR ROBOTS. THE GAME IS TYPICALLY PLAYED IN A KNOCKOUT TOURNAMENT FORMAT,WHERE THEWINNING TEAM ADVANCES TO THE NEXT ROUND.

BOT SPECIFICATIONS:

- EACH ROBOT MUST FIT WITH IN 30CM X 30CM X 30CM (L X B X H) WITH A 5% TOLERANCE.(31.5CM)
- THE MAXIMUM WEIGHT OF THE ROBOT SHOULD NOT EXCEED 5 KG, WITH AN ADDITIONAL 5% TOLERANCE (250 GRAMS) ALLOWED.
- ROBOTS CAN BE EITHER WIRED OR WIRELESS. IF WIRED, THE CABLE MUST BE AT LEAST 4M LONG AND REMAIN LOOSE THROUGHOUT THE GAME.
- ROBOTS MAY USE A MAXIMUM OF 12V DC AT ANY GIVEN TIME. ONLY BATTERY-OPERATED ROBOTS ARE PERMITTED. BATTERY VOLTAGE WILL BE VERIFIED AT THE MOTORS BEFORE THE EVENT.
- ONLY ELECTRIC MOTORS ARE ALLOWED; THE USE OF IC ENGINES OR COMPRESSED AIR MECHANISMS IS STRICTLY PROHIBITED.
- PRE-BUILT RC CARS OR READY-MADE CHASSIS ARE NOT ALLOWED. THE ROBOT MUST BE DESIGNED AND ASSEMBLED BY THE PARTICIPATING TEAM.
- ROBOTS MAY INCLUDE MECHANISMS TO PUSH OR FLICK THE BALL, BUT SUCTION OR ADHESION-BASED METHODS FOR HOLDING THE BALL ARE NOTPERMITTED.
- EACH ROBOT MUST PASS A TECHNICAL INSPECTION AT LEAST 30 MINUTES BEFORE ITS MATCH TO ENSURE COMPLIANCE WITH ALL SPECIFICATIONS.

ARENA SPECIFICATIONS:

- THE GAME IS PLAYED ON A RECTANGULAR FIELD (E.G., 8FT X 4FT).
- GOALS ARE 30CM WIDE ON BOTH ENDS. A LIGHT WEIGHT TENNIS
- BALL IS TYPICALLY USED AS THE GAME BAL

MATCH FORMAT:

- THE COMPETITIONIS CONDUCTED IN A KNOCKOUT FORMAT.
- MATCHES HAVE TIME-LIMITED HALVES, WITH ELIMINATORS, SEMI- FINALS, AND FINALS HAVING DIFFERENT TIME DURATIONS.
- TIEBREAKERS INVOLVE SUDDEN-DEATH OR ADDITIONAL OVERTIME ROUNDS.

ROBO SOCCER

GAMEPLAY RULES:

- THE MATCH BEGINS WITH ROBOTS POSITIONED IN DESIGNATED ZONES AND A REFEREE SIGNALING THE START.
- TEAMS MUST MANEUVER THE BALL USING THEIR ROBOTS WHILE AVOIDING RULE VIOLATIONS.
- ROBOTS MAY BLOCK OPPONENTS, BUT DAMAGING OR INTERFERING WITH CONTROL SYSTEMS IS NOT ALLOWED.
- A GOAL IS COUNTED ONLY IF THE BALL FULLY CROSSES THE GOAL LINE.
- AFTER SCORING, THE GAME RESTARTS FROM THE CENTER.
- TECHNICAL TIMEOUTS ARE ALLOWED FOR MINOR REPAIRS (LIMITED TO 1 MINUTE PER GAME).
- ORGANIZERS MAY MODIFY RULES ON SPOT.

JUDGING & FAIR PLAY:

- THE TEAM WITH THE HIGHEST NUMBER OF GOALS AT THE END OF THE MATCH WINS.
- IN CASE OF A TIE, EXTRA TIME OR PENALTY-BASED DECISIONS MAY BE USED.
- THE REFEREE'S DECISIONS ON GOALS, PENALTIES, AND MATCH DURATION ARE FINAL.

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STUDENT COORDINATOR

ROBO-SUMO

OBJECTIVE:

THE GOAL OF ROBO-SUMO IS TO DESIGN AND BUILD AN AUTONOMOUS ORMANUALLY CONTROLLED ROBOT THAT CAN COMPETE IN A SUMO-STYLE MATCH BY PUSHING THE OPPONENT'S ROBOT OUT OF THE CIRCULAR ARENA WHILE ADHERING TO THE SPECIFIED RULES AND CONSTRAINTS.

BOT SPECIFICATIONS:

- CONTROL TYPE:ROBOTSCAN BE WIRED OR WIRELESS.
- MAXIMUM ONBOARD WEIGHT OF BOT SHOULD NOT EXCEED 5 KG (ADDITIONAL 5% TOLERANCE I.E. 250 GRAMS).
- SIZE CONSTRAINT: THE MAXIMUM DIMENSIONS ALLOWED ARE 30 X 30 X 30 CM (WITH A 5% TOLERANCE I.E. 31.5CM).
- CONSTRUCTION RULE: ROBOTS MUST BE CUSTOM-BUILT— PRE-ASSEMBLED KITS, LEGO PARTS, OR READY-MADE COMPONENTS ARE NOT PERMITTED.

POWER SOURCE:

- THE ROBOT MUST BE ELECTRICALLY POWERED (NO IC ENGINES ALLOWED).
- IT MAY USE A BATTERY FIXED ON THE ROBOT OR A STATIONARY POWER SOURCE CONNECTED VIA A CORD.

VOLTAGE RESTRICTION:

- THE MAXIMUM POTENTIAL DIFFERENCE BETWEEN ANY TWO POINTS ON THE BOT MUST NOT EXCEED 12.5V.

BACKUP REQUIREMENT:

- TEAMS MUST HAVE SPARE BATTERIES TO AVOID DISQUALIFICATION IF THE ROBOT BECOMES NON-FUNCTIONAL DURING A MATCH.

WIRE LENGTH (FOR WIRED BOTS):

- THE WIRE MUST BE LONG ENOUGH TO COVER THE ARENA AND SHOULD REMAIN SLACK THROUGHOUT THE MATCH.

ARENA SPECIFICATIONS:

- THE ARENA FEATURES A CIRCULAR PLYWOOD PLATFORM WITH A DIAMETER OF 2 METERS.
- A WHITE BORDER WILL ASSIST SUMO ROBOTS IN NAVIGATION.
- THE PLATFORM IS SLIGHTLY ELEVATED FROM THE BASE TO CLEARLY INDICATE WHEN A ROBOT HAS BEEN PUSHED OUT.

ROBO-SUMO

RULES AND REGULATIONS:

- TEAM COMPOSITION:EACH TEAM MUST HAVE 2 TO 4 MEMBERS, AND PARTICIPATION IS OPEN ONLY TO CURRENTLY ENROLLED COLLEGE

STUDENTS.

- MATCH FORMAT:
- EACH MATCH CONSISTS OF TWO ROUNDS, 1.5 MINUTES EACH. IN CASE OF A TIE, AN ADDITIONAL 1-MINUTE ROUND WILL BE CONDUCTED.

SCORING SYSTEM:

- WINNING TEAM GETS 1 POINTS, WHILE THE LOSING TEAM GETS 0 POINTS.
- TO WIN, A ROBOT MUST PUSH THE OPPONENT COMPLETELY OUT OF THE ARENA.
- A SUMO-BOT IS NOT CONSIDERED OUT UNTIL IT FULLY FALLS OFF THE PLATFORM. HANGING OVER THE EDGE DOES NOT DISQUALIFY A SUMO-BOT.

POINT ALLOCATION:

- THE ROBOT REMAINING ON THE ARENA AT THE END OF THE ROUND IS AWARDED THE POINT.
- IF BOTH ROBOTS FALL, THE ONE THAT TOUCHES THE GROUND LAST EARNS THE POINT.
- IF A ROBOT BECOMES DISABLED OR IMMOBILE, THE OPPONENT WINS THE POINT. THE ROBOT WITH THE MOST POINTS WINS THE MATCH (FINAL SCORING DETAILS WILL BE ANNOUNCED DURING THE EVENT).
- IF NO WINNER IS DECIDED WITHIN THE GIVEN TIME, A 2-MINUTE TIEBREAKER MATCH (DO-OR-DIE) WILL TAKE PLACE, WHERE THE FIRST POINT WINS THE MATCH.

RESTART CONDITIONS:

- IF ROBOTS REMAIN ENTANGLED FOR MORE THAN 10 SECONDS, THE JUDGE WILL RESTART THE MATCH FROM THE INITIAL POSITION.
- ENTANGLEMENT IS DEFINED AS ROBOTS LOCKED TOGETHER WITHOUT MAKING SIGNIFICANT PROGRESS TOWARD THE ARENA'S EDGE.

TIME-OUTS & WEIGHT CONSIDERATIONS:

- ONE TECHNICAL TIMEOUT (60 SECONDS) IS ALLOWED PER TEAM PER MATCH.

CONDUCT & FAIR PLAY:

- PARTICIPANTS MUST FOLLOW THE RULES AND JUDGES' DECISIONS.
- MISBEHAVIOR MAY LEAD TO DISQUALIFICATION FROM THE COMPETITION.
- ROBOTS OR PARTICIPANTS ATTEMPTING DELIBERATE INTERFERENCE WITH OPPONENTS OR DAMAGING THE ARENA WILL BE DISQUALIFIED.

AUTHORITY OF ORGANIZERS:

- THE EVENT COORDINATORS HAVE THE RIGHT TO MODIFY RULES ON THE SPOT IF NECESSARY.
- ALL DECISIONS REGARDING GAMEPLAY, SCORING, AND PARTICIPATION MADE BY THE ORGANIZING COMMITTEE ARE FINAL.
- MATCH CONTROL:
- THE MATCH CAN BE STOPPED, RESUMED, OR ENDED ONLY BY A JUDGE'S ANNOUNCEMENT.

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STUDENT COORDINATOR

LINE FOLLOWER

OBJECTIVE:

PARTICIPANTS MUST DESIGN ANAUTONOMOUS ROBOT CAPABLE OF DETECTING AND FOLLOWING A DESIGNATED PATH—EITHER BLACK ON A WHITE BACKGROUND OR WHITE ON A BLACKGROUND—TO TRAVEL FROM THE STARTING POINT TO THE FINISH LINE IN THE SHORTEST POSSIBLE TIME.

GAMEPLAY :

- ROBOTS WILL BE POSITIONED AT THE STARTING POINT, AND THE TIMER STARTS WHEN THE RUN BEGINS.
- THE TIMER WILL CONTINUE UNTIL THE ROBOT REACHES THE FINISH LINE.
- EACH TEAM IS ALLOWED ONE TRIALS PER ROUND THAT TO CALIBRATION.

A ROBOT MUST RESTART IF:

- IT FAILS TO START WITHIN ONE MINUTE AFTER PRESSING THE START BUTTON.
- IT IS TOUCHED BY A HUMAN (5-SECOND PENALTY PER TOUCH). IT MOVES OFF THE TRACK.
- THEN REFEREE ORDERS A RESTART.

CHECK POINTS WILL BE PLACED ALONG THE TRACK:

- IF THE ROBOT FAILS TO REACH A CHECKPOINT, IT MUST RETURN TO THE PREVIOUS CHECKPOINT.
- IF THE ROBOT FAILS AGAIN, THE PARTICIPANT HAS THE OPTION TO SKIP THE CHECKPOINT AT THE COST OF A 30-SECOND PENALTY.
- SCORING, PENALTIES, AND TIME LIMITS WILL BE ANNOUNCED ON THE DAY OF THE EVENT.

TRACK SPECIFICATIONS:

- THE TRACK WILL BE MADE OF A FLEXIBLE SHEET WITH CONTRASTING LINES: EITHER BLACK LINES ON A WHITE BACKGROUND OR WHITE LINES ON A BLACK BACKGROUND.
- THE COURSE WILL FEATURE VARIOUS ANGLES, INCLUDING ACUTE, OBTUSE, AND RIGHT ANGLES, AS WELL AS CURVES.
- DISCONTINUITIES MAY BE PRESENT AT CERTAIN POINTS ALONG THE TRACK.
- CHECKPOINTS WILL BE PLACED AT KEY LOCATIONS TO ENSURE PROGRESS TRACKING.
- THE TRACK LAYOUT WILL BE REVEALED ON THE DAY OF THE EVENT.

LINE FOLLOWER

ROBOT SPECIFICATIONS:

- THE ROBOT MUST OPERATE COMPLETELY AUTONOMOUSLY, WITHOUT HUMAN INTERVENTION.
- MAXIMUM DIMENSIONS (INCLUDING TOLERANCE):
  WIDTH: 280MM
  LENGTH: 280MM
  HEIGHT: 200MM
- MAXIMUM WEIGHT: 3 KG.
- THE ROBOT MUST BE STARTED MANUALLY USING A START BUTTON.
- LEGO PARTS AND PRE-ASSEMBLED ROBOTS ARE STRICTLY PROHIBITED.
- THE MAXIMUM VOLTAGE BETWEEN ANY TWO POINTS ON THE ROBOT MUST NOT EXCEED 12V.
- THE ROBOT CAN BE POWERED BY A 12V BATTERY MOUNTED ON THE BOT OR A STATIONARY POWER SOURCE CONNECTED THROUGH A CORD.
- SPARE BATTERIES MUST BE CARRIED BY PARTICIPANTS TO AVOID DISQUALIFICATION DUE TO POWER FAILURE DURING A TRIAL.

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STUDENT COORDINATOR

MAZE RACING

OBJECTIVE:

GET READY TO NAVIGATE! STEP INTO THE FUTURE OF ROBOTICS AT OUR THRILLING ROBOT MAZE SOLVER EVENT! PARTICIPANTS WILL HAVE THE OPPORTUNITY TO BUILD AND PROGRAM ROBOTS TO NAVIGATE COMPLEX MAZES. THE ROBOT MUST NAVIGATE THE MAZE AND SELECT THE CORRECT PATH TO REACH THE GOAL. THE FIRST ROBOT TO ARRIVE AT THE FINISH.

MAZE ROBOT SPECIFICATIONS:

- DESIGN AND BUILD A BOT THAT CAN OVERCOME VARIOUS CHALLENGES SUCH AS MAZE AND OBSTACLES.
- THE ROBOT CAN BE AUTONOMOUS OR MANUALLY CONTROLLABLE. MAXIMUM DIMENSIONS (INCLUDING TOLERANCE):
  WIDTH: 17CM LENGTH: 17CM HEIGHT: 12CM
- THE WIDTH OF PATH WILL BE 20CM AND THE OBSTACLES WILL BE PLACED 20CM APART.
- THE WIRELESS BOT MUST BE CONTROLLED MANUALLY VIA WIRELESS COMMUNICATION (EX: BLUETOOTH, RADIO, WI-FI).
- THE ROBOT MUST NOT BE MADE FROM LEGO PARTS, OR ANY READY-MADE ASSEMBLY KITS, AND IT MUST BE POWERED ELECTRICALLY, USE OF IC ENGINE IS NOT ALLOWED.
- DIMENSIONS WILL BE STRICTLY FOLLOWED, IF NOT THE TEAM WILL BE DISQUALIFIED.
- POTENTIAL BETWEEN ANY TWO POINTS ON THE ROBOT SHOULD NOT EXCEED 12V.
- SPARE BATTERIES MUST BE CARRIED BY PARTICIPANTS TO AVOID DISQUALIFICATION DUE TO POWER FAILURE DURING A TRIAL.
- IF THE BOT REACHES THE WRONG WAY / PATH PENALTY TIME WILL BE ADDED TO THE FINAL TRACK COMPLETION TIME.
- IF THE BOT REACHES THE WRONG WAY / PATH PENALTY TIME WILL BE ADDED TO THE FINAL TRACK COMPLETION TIME.

MAZE RACING

GAMEPLAY & SCORING:

THE ROBOT WILL BE PLACED AT START LINE. THE ROBOT CAN START THE LAP WHEN TIMER STARTS. IF THE BOT TOUCHES THE OBSTACLES , PENALTY TIME WILL BE ADDED TO THE FINAL TRACK COMPLETION TIME. THERE ARE PENALTIES FOR EACH WRONG OBSTACLE, PENALTIES WILL BE ADDED TO TOTAL TIME. TECHNICAL TIME OUT OF 2 MINS WILL BE PROVIDED. TIMER WILL STOP AS SOON AS ROBOT CROSSES FINISH LINE. SCORING AND PENALTIES RULES WILL BE ANNOUNCED ON THE DAY OF EVENT. ONLY 2 MEMBERS WILL BE ALLOWED INSIDE THE ALLOTTED ROOM OF MAZE SOLVER EVENT NO MOBILE PHONES ARE ALLOWED.

EVENT GUIDELINES:

- TEAM OF MINIMUM 2 AND MAXIMUM 4 MEMBERS IS ALLOWED. ONLY 2 MEMBERS WILL BE ALLOWED INSIDE THE ALLOTTED ROOM OF MAZE SOLVER EVENT.
- MOBILE PHONES ARE NOTALLOWED

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STUDENT COORDINATOR

THE JUDGES' DECISION WILL BE FINAL AT ALL TIMES. THE EVENT SCHEDULE AND RULES MAY BE MODIFIED AS NECESSARY.